

SIT292 Module 2 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	Review
Question 1	4/4	
Question 2	1/1	
Question 3	2/2	
Question 4	2/2	
Question 5	1/1	
Total	10/10	(100%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 2 Self-Assessment
Session ID:	5031708919757769
Exam Start:	Sun Jul 20 2025 13:31:29
Exam Stop:	Sun Jul 20 2025 14:12:13
Time Spent:	0:40:44

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<https://www.newcastle.ac.uk>).

Question 1

a)

Calculate the following (use the controls at the top to set the size of the matrix)

$$-4 \begin{pmatrix} -3 & 4 & -3 \\ -5 & 4 & 5 \end{pmatrix} - 3 \begin{pmatrix} -2 & -3 & -1 \\ 4 & 1 & 5 \end{pmatrix} =$$

Rows: Columns:

$$\begin{pmatrix} \boxed{18} & \boxed{-7} & \boxed{15} \\ \boxed{8} & \boxed{-19} & \boxed{-35} \end{pmatrix} \quad \checkmark$$

Expected answer: $\begin{pmatrix} 18 & -7 & 15 \\ 8 & -19 & -35 \end{pmatrix}$

Score: 2/2 ✓

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b)

Calculate the following

$$\begin{pmatrix} 4 & -2 \\ -3 & 1 \\ -4 & 1 \\ 2 & 3 \end{pmatrix}^T =$$

Rows: Columns:

$$\begin{pmatrix} \boxed{4} & \boxed{-3} & \boxed{-4} & \boxed{2} \\ \boxed{-2} & \boxed{1} & \boxed{1} & \boxed{3} \end{pmatrix} \checkmark$$

Expected answer: $\begin{pmatrix} 4 & -3 & -4 & 2 \\ -2 & 1 & 1 & 3 \end{pmatrix}$

Score: 2/2

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Your answer is correct. You were awarded 2 marks.

You scored 2 marks for this part.

Question 2

Let $A = \begin{pmatrix} -3 & -1 & -1 & 0 & -3 \\ -1 & 1 & -2 & -3 & -1 \end{pmatrix}$

The matrix transformation of A is $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ for which numbers n and m ?

$n =$

Expected answer:

and $m =$

Expected answer:

m

✓ Your answer is correct. You were awarded **0.5** marks.

n

✓ Your answer is correct. You were awarded **0.5** marks.You scored **1** mark for this part.

Question 3

Let $A = \begin{pmatrix} -2 & 1 & 0 \\ -2 & 2 & 0 \\ 0 & -3 & 4 \end{pmatrix}$

For a general vector $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, determine $T_A(x)$. Input x_1 as x_1 or x_1 , etc. Use the same notation for all variables, don't mix it.

$$T_A(x) = \begin{pmatrix} \boxed{-2x_1 + x_2} & -2x_1 + x_2 \quad \checkmark \\ \text{Expected answer: } \underline{-2x_1 + x_2} & -2x_1 + x_2 \\ \boxed{-2x_1 + 2x_2} & -2x_1 + 2x_2 \quad \checkmark \\ \text{Expected answer: } \underline{-2x_1 + 2x_2} & -2x_1 + 2x_2 \\ \boxed{-3x_2 + 4x_3} & -3x_2 + 4x_3 \quad \checkmark \\ \text{Expected answer: } \underline{-3x_2 + 4x_3} & -3x_2 + 4x_3 \end{pmatrix}$$

Score: 2/2 ✓

✓ Your answer is correct. You were awarded 2 marks.

You scored 2 marks for this part.

Question 4

Let $A = \begin{pmatrix} 2 & -2 & 2 & 3 \\ 3 & -1 & 5 & -4 \end{pmatrix}$, $B = \begin{pmatrix} 4 & -3 & 1 \\ 5 & -4 & 2 \\ -4 & -4 & -2 \\ 2 & 0 & 1 \end{pmatrix}$.

Compute $AB =$ Rows: Columns:

$$\begin{pmatrix} -4 & -6 & -3 \\ -21 & -25 & -13 \end{pmatrix}$$



Expected answer: $\begin{pmatrix} -4 & -6 & -3 \\ -21 & -25 & -13 \end{pmatrix}$

Score: 2/2 ✓

✓ Your answer is correct. You were awarded 2 marks.

You scored 2 marks for this part.

Question 5

Consider the following nutrition table.

	Fat	Sodium	Carbohydrates
Banana	0.2 g	1.2mg	27 g
Cornflakes (bowl)	0.3g	236 mg	24g
Milk (cup)	3 g	110.9mg	1.2 g

The information in the table is represented as a matrix A, i.e.,

$$A = \begin{pmatrix} 0.2 & 1.2 & 27 \\ 0.3 & 236 & 24 \\ 3 & 110.9 & 16 \end{pmatrix}$$

Suppose we have a points system that scores 0.9 points per gram of fat, 0.8 per mg of sodium, and 2 per gram of carbohydrates. Which of the following will give the total points scored by a banana, bowl of cornflakes, and cup of milk?

- ☐ $(0.9 \ 0.8 \ 2) A$
☐ $\begin{pmatrix} 0.9 \\ 0.8 \\ 2 \end{pmatrix} A$
☒ $A \begin{pmatrix} 0.9 \\ 0.8 \\ 2 \end{pmatrix}$
☐ $A (0.9 \ 0.8 \ 2)$



Expected answer:

- ☐ $(0.9 \ 0.8 \ 2) A$
☐ $\begin{pmatrix} 0.9 \\ 0.8 \\ 2 \end{pmatrix} A$
☒ $A \begin{pmatrix} 0.9 \\ 0.8 \\ 2 \end{pmatrix}$
☐ $A (0.9 \ 0.8 \ 2)$

Score: 1/1 ✓

- ✓ You chose a correct answer. You were awarded **1** mark.
You scored **1** mark for this part.

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Self Assessment 2

mult by scalar first, +3 and subtract the result

$$-4 \begin{pmatrix} -3 & 4 & -3 \\ -5 & 4 & 5 \end{pmatrix} - 3 \begin{pmatrix} -2 & -3 & -1 \\ 4 & 1 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 12 & -16 & 12 \\ 20 & -16 & -20 \end{pmatrix} + \begin{pmatrix} -6 & -9 & -3 \\ -12 & -3 & -15 \end{pmatrix} = \begin{vmatrix} 18 & -7 & 15 \\ 8 & -19 & -35 \end{vmatrix}$$

5x2

$$A = \begin{pmatrix} -3 & -1 & -1 & 0 & -3 \\ -1 & 1 & -2 & -3 & -1 \end{pmatrix} \quad A: T_A: \mathbb{R}^{ids} \rightarrow \mathbb{R}^{rows}$$

$$A = \begin{pmatrix} -2 & 1 & 0 \\ -2 & 2 & 0 \\ 0 & -3 & 4 \end{pmatrix} \quad T_A(x) = \begin{vmatrix} -2x_1 + 1x_2 + 0x_3 \\ -2x_1 + 2x_2 + 0x_3 \\ 0x_1 - 3x_2 + 4x_3 \end{vmatrix}$$

$$= \begin{vmatrix} -2x_1 + x_2 \\ -2x_1 + 2x_2 \\ -3x_2 + 4x_3 \end{vmatrix}$$

$$A = \begin{pmatrix} 2 & -2 & 2 & 3 \\ 3 & -1 & 5 & -4 \end{pmatrix} \quad B = \begin{pmatrix} 4 & -3 & 1 \\ 5 & -4 & 2 \\ -4 & -4 & -2 \\ 2 & 0 & 1 \end{pmatrix}$$

$$AB = \begin{vmatrix} -4 & -6 & -8 \\ -21 & -25 & -13 \end{vmatrix}$$

$(2 \times 4) + (-2 \times -3) + (2 \times 1) + (3 \times 4)$
 $(3 \times 4) + (-1 \times -3) + (5 \times 1) + (-4 \times 4)$
 $(-2 \times -4) + (-2 \times -4) + (2 \times -2) + (3 \times -4)$
 $(-1 \times -4) + (-1 \times -4) + (-2 \times -2) + (-4 \times -4)$
 $(5 \times -2) + (5 \times -4) + (-2 \times -2) + (-2 \times -4)$
 $(-4 \times -2) + (-4 \times -4) + (-2 \times -2) + (-4 \times -4)$

Number

$$A = \begin{pmatrix} 0.2 & 1.2 & 27 \\ 0.3 & 236 & 24 \\ 3 & 110.9 & 16 \end{pmatrix} \begin{pmatrix} 0.9 \text{ points} \\ 0.8 \text{ points} \\ 2 \end{pmatrix}$$

eg. $R1 = (0.2 \times 0.9) + (1.2 \times 0.8) + (27 \times 2)$

Content for Banana

Other way doesn't work

Summary

Overall I'm good friends with matrix multiplication, so I was able to get through this without any major obstacles. I found the problems challenging enough to be engaging but not so much that I was overwhelmed, very happy with the outcome. No calculator required at any steps this time since it all comes back to integer multiplication!